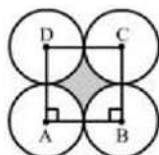




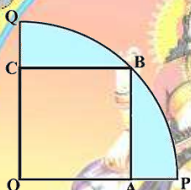
- How long will a boy take to walk to 3km/hr once around a circular park of radius 210m.
(a) 26 min and 24 sec (b) 16 min and 24 sec
(c) 6 min and 20 sec (d) 8 min and 20 sec
- Four equal circles are described about the four corners of a square so that each touches two of the others, as shown in the figure. Find the area of the shaded region, each side of the square measuring 14 cm.

$$\left[\text{Use } \pi = \frac{22}{7} \right]$$

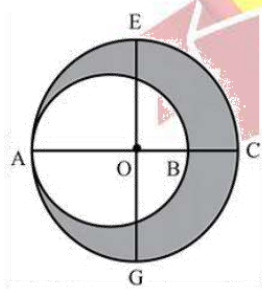


- (a) 42 sq. cm (b) 24 sq. cm
(c) 12 sq. cm (d) 43 sq. cm
- A square OABC is inscribed in a quadrant OPBQ of a circle as shown in the figure. If OA = 14 cm, find the area of the shaded region

$$\left[\text{Use } \pi = \frac{22}{7} \right]$$



- (a) 110 cm² (b) 112 cm²
(c) 113 cm² (d) 114 cm²
- In the given figure, O is the centre of the bigger circle and AC is its diameter. Another circle with AB as diameter is drawn. If AC = 54 cm and BC = 10 cm, the area of the region is ----



- (a) 70 cm² (b) 77 cm²
(c) 770 cm² (d) 385 cm²
- f(x) is called p.d.f of the continuous random variable, if
(a) $\int_0^\infty f(x) dx = 1$ (b) $\int_{-\infty}^0 f(x) dx = 1$
(c) $\int_{-\infty}^\infty f(x) dx = 1$ (d) $\int_{-a}^a f(x) dx = 1$
- If the probability distribution function of a random variable X is given by

$$f(x) = \frac{1}{7}, \quad \forall 1 < x < 8$$

$$= 0, \quad \text{otherwise}$$

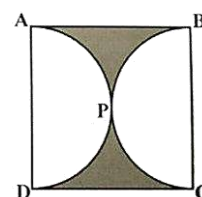
Then the value of $P(3 < X < 10)$ is

- (a) 6/7 (b) 5/7
(c) 3/7 (d) 2/7

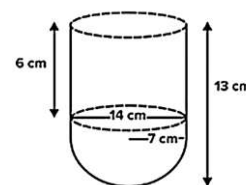
- Given the $X \sim B(n = 10, p)$ and $E(X) = 8$, the value of p is
(a) 8 (b) 0.8
(c) 0.88 (d) 0.08
- In a binomial distribution, mean and variance are 12 and 4 respectively. the parameters of the distribution are
(a) $n = 18$ and $p = 2/3$ (b) $n = 9$ and $p = 12/13$
(c) $n = 28$ and $p = 3/2$ (d) $n = 8$ and $p = 3/4$
- A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Tanmay wins if all the tosses give the same result, i.e. three heads or three tails, and loses otherwise. Calculate the probability that Tanmay will lose the game.

- (a) 3/4 (b) 1/4
(c) 2/3 (d) 2/4
- A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$. Find the number of blue balls in the jar.

- (a) 10 (b) 8
(c) 18 (d) 28
- Find the area of the shaded region as shown in the figure, if ABCD is a square of side 14 cm and APD and BPC are semicircles.



- (a) 24 cm² (b) 42 cm²
(c) 12 cm² (d) 21 cm²
- A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.



- (a) 572 cm² (b) 672 cm²
(c) 472 cm² (d) 275 cm²
- A and B start a business with initial investments in the ratio 5:7. After 6 months, A withdraws half his investment, and B doubles his investment. If at the end of the year, the profit is ₹24,000, what is A's share of the profit?
(a) ₹9,000 (b) ₹7,500
(c) ₹10,000 (d) ₹8,000



14. A car travels from A to B at a speed of x km/h and returns from B to A at a speed of y km/h. If the average speed for the entire is 48 km/h and the distance between A and B is 96 km, what is the value of $x+y$?
- (a) 96 km/hr (b) 120 km/hr
(c) 144 km/hr (d) 192 km/hr
15. If R is a relation defined on the set Z of integers by the rule $(x, y) \in R \Leftrightarrow x^2 + y^2 = 9$, then the domain of R is
- (a) $\{2, 0, -2\}$ (b) $\{-3, 0, 3\}$
(c) $\{4, 2, 3\}$ (d) $\{3, 2, -1\}$
16. The surface area of a balloon being inflated changes at a constant rate. If initially its radius is 4 units and after 3 seconds it is 6 units. What is the radius after t seconds?
- (a) $\sqrt{10t + 9}$ (b) $\sqrt{\frac{20}{3}t + 16}$
(c) $\sqrt{8t + 16}$ (d) $\sqrt{\frac{15}{2}t + 9}$
17. A physicist is studying the relationship between the magnetic field strength x (in Tesla) and the electric field strength y (in volts per meter) at a particular point in space. The interaction between these fields uses an equation: $(x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$. At a certain moment, the magnetic field strength is $x=2$ Tesla and the electric field strength is $y=1$ volt per meter. Which of the following equations best describes the relationship between the magnetic and electric field strengths at this point?
- (a) $y = \frac{x^2}{x^2+1}$ (b) $y = 0.25x^2$
(c) $y = \frac{2x}{x+1}$ (d) $y = \frac{x^2}{x^3+3}$
18. A family of circles lies in the second quadrant and touches both the x -axis and y -axis. The general form of the equation for such a circle is: $(x+a)^2 + (y-a)^2 = a^2, a > 0$. Which of the following represent the differential equation for this family of circles?
- (a) $\frac{dy}{dx} + \frac{y}{x} = 0$
(b) $x^2 + y^2 + 2ax - 2ay = 0$
(c) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - \frac{1}{y} = 0$
(d) $\frac{dy}{dx} = -\frac{y}{x}$
19. Solve $\frac{dy}{dx} + y = 1$ and select the correct answer.
- (a) $y = 1 + Ce^{-x}$ (b) $y = Ce^x - 1$
(c) $y = Ce^{-x}$ (d) $y = 1 - Ce^x$
20. In a triangle ABC , $A = 60^\circ, B = 45^\circ$ and side $a = 10$ calculate the length of side b using the Law of Sines.
- (a) 15.04 (b) 8.2
(c) 6.7 (d) 10.4
21. The domain of the function $\cos^{-1}(2x - 1)$ is
- (a) $[0, 1]$ (b) $[-1, 1]$
(c) $[-1, 1]$ (d) $[0, \pi]$
22. If $3 \tan^{-1} x + \cot^{-1} x = \pi$ then x equals
- (a) 0 (b) 1
(c) -1 (d) $\frac{1}{3}$
23. The lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x-1}{-2} = \frac{y-2}{-4} = \frac{z-3}{-6}$ are
- (a) Coincident (b) Skew
(c) Intersecting (d) Parallel
24. Two motorcycle running at the speed more than the allowed speed on roads represented by the line $\vec{r} = \lambda(\hat{i} + 2\hat{j} - \hat{k})$ and $\vec{r} = (\hat{3i} + \hat{3j}) + \mu(\hat{2i} + \hat{j} + \hat{k})$ respectively. Find the point at which motorcycle will collide.
- (a) $\vec{r} = \hat{i} + 2\hat{j} - \hat{k}$ (b) $\vec{r} = \hat{3i} + \hat{3j} - \hat{k}$
(c) $\vec{r} = 2\hat{i} + 4\hat{j} - 2\hat{k}$ (d) $\vec{r} = \hat{5i} + \hat{5j} - 2\hat{k}$
25. If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$ then $4x^2 - 4xy + y^2$ is equal to
- (a) $-4 \sin^2 \alpha$ (b) $4 \sin^2 \alpha$
(c) 4 (d) $2 \sin 2\alpha$
26. The equation of ellipse is given $3x^2 + 5y^2 = 32$, with the center at the origin and x -axis as the major axis. Which of the following points lies on the ellipse?
- (a) $(-3, 1)$ and $(2, -2)$
(b) $(3, -2)$ and $(1, -1)$
(c) $(-2, 3)$ and $(2, 2)$
(d) $(3, 1)$ and $(-2, -3)$
27. If the coefficients of 2^{nd} , 3^{rd} and the 4^{th} terms in the expansion of $(1+x)^n$ are in Arithmetic Progression, then value of n is
- (a) 2 (b) 7
(c) 11 (d) 14
28. The number of irrational terms in the expansion of $\left(4^{\frac{1}{5}} + 7^{\frac{1}{10}}\right)^{45}$ is
- (a) 40 (b) 5
(c) 41 (d) 37
29. How many ways can a lawn tennis mixed double be made up from 7 married couples if no husband and wife play in the same time?
- (a) 550 (b) 790
(c) 840 (d) 768
30. How many five-digit telephone numbers can be constructed using the digits 0 to 9. If each number starts with 67 and no digit appears more than once?
- (a) 336 (b) 231
(c) 456 (d) 342





31. Statements:

1. All books are papers.
2. Some papers are pens.

Conclusions:

- I. Some books are pens.
- II. Some pens are papers.

- (a) Only I follows
- (b) Only II follows
- (c) Both I and II follow
- (d) Neither I nor II follows

32. Pointing to a boy, Rahul says, "He is the son of my grandmother's only daughter." How is the boy related to Rahul?

- (a) Brother
- (b) Cousin
- (c) Nephew
- (d) Uncle

33. Find the next number in the series:

2, 6, 12, 20, 30, _____

- (a) 36
- (b) 42
- (c) 44
- (d) 48

34. Five friends - P, Q, R, T, and U - sit in a row facing North.

Q is second to the left of T.

P is to the immediate left of Q.

U is to the immediate right of T.

R is not at an end.

Who sits at the leftmost end?

- (a) P
- (b) Q
- (c) R
- (d) T

35. If DOG is written as 26, CAT as 24, how is HEN written?

- (a) 25
- (b) 27
- (c) 33
- (d) 45

36. Four people - P, Q, R, and S - play a game.

P always scores more than Q.

Q always scores more than S.

R scores less than P but more than Q.

Who scores the second highest?

- (a) P
- (b) Q
- (c) R
- (d) S

37. Which number does not belong to the group?

- A. 121
- B. 144
- C. 169
- D. 225

- (a) 121
- (b) 144
- (c) 169
- (d) 225

38. Climate scientists argue that planting trees in urban areas the line evidence showing that trees absorb harmful gases and particulate matter. However, some critics argue that urban tree planting programs are expensive and do not address the root causes of pollution, such as industrial emissions.

Which of the following is a logical inference from the passage?

- (a) Trees are more effective at reducing air pollution than industrial controls.
- (b) Urban tree planting programs are not feasible in industrial cities.
- (c) Tree planting programs can complement other anti-pollution measures.
- (d) Planting trees has no significant impact on urban pollution.

39. Climate scientists argue that planting trees in urban areas can significantly reduce air pollution levels. They cite evidence showing that trees absorb harmful gases and particulate matter. However, some critics argue that urban tree planting programs are expensive and do not address the root causes of pollution, such as industrial emissions.

Which of the following strengthens the argument in favour of urban tree planting programs?

- (a) Urban trees improve mental health for city residents
- (b) Studies show that cities with more trees have cleaner air.
- (c) Tree planting programs require significant funding and maintenance
- (d) Urban tree planting programs can only be implemented in wealthy areas.

40. In a certain code "CLOUD" is written as "ENQWF". How "CRACK" is written in that code?

- (a) SPUPSR
- (b) ETCCEM
- (c) ESAPTO
- (d) NJELVP

41. Find the next number in the sequence: 2, 4, 8, 16, 32.?

- (a) 48
- (b) 64
- (c) 40
- (d) 128

42. Complete the analogy: Book is to Read as Knife is to ____.

- (a) Eat
- (b) Cut
- (c) Sharpen
- (d) Clean

43. In a small town, there are types of vehicles: cars, bikes, and buses. The town council recently conducted a survey to understand the usage of these vehicles. The data showed that 50% of the population owns cars, 30% own bikes, and 20% use buses as their primary mode of transportation. Additionally the survey revealed that 40% of car owners also own bikes, while only 10% of bike owners also use buses. The council plans to introduce incentives to promote bus usage.

What can be inferred about the town's population based on the survey?

- (a) Most of the population owns bikes and uses buses.
- (b) Cars are the most preferred mode of transportation.
- (c) Bike owners are less likely to use buses
- (d) The population prefers buses over cars and bikes.





44. In a small town, there are types of vehicles: cars, bikes, and buses. The town council recently conducted a survey to understand the usage of these vehicles. The data showed that 50% of the population owns cars, 30% own bikes, and 20% use buses as their primary mode of transportation. Additionally the survey revealed that 40% of car owners also own bikes, while only 10% of bike owners also use buses. The council plans to introduce incentives to promote bus usage.

If the council introduces a policy offering free bus passes to 30% of the population, and all these individuals start using buses exclusively. what percentage of the population would now rely on buses?

- (a)30% (b)40%
(c)50% (d)60%

45. A factory produces 1200 units of a product in 6 hours with 8 machines working at the same rate. If 10 machines work at the same rate. how many units will be produced in 5 hours?

- (a)1000 (b)1250
(c)1500 (d)1600

46. A person walks 5 km north, then turns right and walks 3 km. Then, he turns right again and walks 5 km. Finally, he turns left and walks 2 km. In which direction is he from his starting point?

- (a)North (b)South
(c)East (d)West

47. Given below is a statement and two assumptions. These assumptions may or may not be implicit in the statement.

Statement:

The company announced a 20% salary hike for all its employees.

Assumption A:The company is making good profits.

Assumption B:Employees are satisfied with the hike.

Determine whether one or both assumptions are implicit or not in the given statement and choose the correct option from the options given below:

- (a)Only Assumption A is implicit
(b)Only Assumption B is implicit
(c)Neither Assumption A nor Assumption B are implicit
(d)Both Assumption A and Assumption B are implicit

48. If January 1, 2025, is a Wednesday, what will be the day of the week on January 1, 2026?

- (a)Wednesday (b)Thursday
(c)Friday (d)Saturday

49. Given below are two statements, one is labelled as Assertion (A) and other one labelled as Reason (R).

Assertion (A): All squares are rectangles.

Reason (R): A rectangle is a quadrilateral with opposite sides equal and parallel, and a square satisfies these conditions.

In light of the above statements, choose the correct answer from the options given below.

- (a)Both (A) and (R) are true and (R) is the correct explanation of (A)
(b)Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(c)(A) is true but (R) is false
(d)(A) is false but (R) is true

50. Which word does not belong to the group?

- (a)Dog (b)Cat
(c)Bird (d)Car

51. Statement: "A new smartphone has been launched with features that no other phone currently has."

Assumption:

- I. People are always interested in smartphones with unique features.
II. No other phone will have similar features in the future

- (a)Only I is implicit
(b)Only II is implicit
(c)Both I and II are implicit
(d)Neither I nor II is implicit

52. If the following statements are true:

1. All apples are fruits.
2. Some fruits are sweet.

Which of the following conclusions logically follows?

- (a)All apples are sweet.
(b)Some fruits are not apples.
(c)Some apples are sweet.
(d)All fruits are sweet

53. Four friends (P, Q, R and S) are seated around a circular table.

1. P is to the immediate left of Q.
2. R is opposite Q.
3. S is not next to Q.

Who is to the immediate right of S?

- (a)P (b)Q
(c)R (d)Cannot be determined

54. Which of the following groups can be represented by three overlapping circles in a Venn Diagram?

- (a) Birds, Mammals, Fish
(b) Doctor, Engineers, Teachers
(c) Men, Father, Athletes
(d) Cats, Dogs, Lions





55. Identify the odd one out:

- (a) 2, 3, 5, 7 (b) 11, 13, 17, 19
(c) 23, 29, 31, 33 (d) 37, 41, 43, 47

56. If January 1, 2024, is a Monday, what day will January 1, 2025, be?

- (a) Tuesday (b) Wednesday
(c) Thursday (d) Friday

57. Find the missing number:

2, 6, 12, 20, 30, ?, 56

- (a) 38 (b) 40
(c) 42 (d) 44

58. If TABLE is coded as XEFVI, how is CHAIR coded?

- (a) FKDLU (b) GLESV
(c) FKGLU (d) DKGLU

59. A is mother of B. B is the sister of C. C is the father of D. What is A's relation to D?

- (a) Grandmother (b) Mother
(c) Aunt (d) Cousin

60. A, B, C, D, and E are sitting in a row.

1. C is between B and D.
2. A is to the left of B but right of E.
Who is sitting at the center?

- (a) B (b) C (c) D (d) A

61. Read the following short passage carefully and answer the following question.

The avowed pupose of the exact sciences is to establish complete intellectual control over experience in terms of precise rules which can be formally set out and empirically tested. Could that ideal be fully achieved, all truth and all error could henceforth be ascribed to an exact theory of the universe, while we who accept this theory would be relieved of any occasion for exercising our personal judgement. We should only have to follow the rules faithfully. Classical mechanics approaches this ideal so closely that it is often thought to have achieved it. But this leaves out of account the element of personal judgement involved in aplying the formulae of mechanics to the fact of experience.

The purpose of the exact sciences is to

- (a) Form opinions about our experience
(b) Formulate principles which will help us to exercise our personal judgement
(c) Assert our intellectual superiority
(d) Make formal and testable rules which can help verify experience.

62. Choose the correct word which can be substituted for the given words/sentence.

To examine ones's own thoughts and feelings.

- (a) Restrospection (b) Introspection
(c) Meditation (d) Reflection

63. Find the best fit into the blank space of the following sentence.

I hate sitting him as he always smells to garlic.

- (a) Besides (b) Along
(c) At (d) Beside

64. Read the passage carefully and answer the following question.

Certain words/phrases in the passage are ITALICISED to help you to locate them whide answering the question

The present industrial age is steadily placing into out hands power and instruments which are expected to relieve us of mechanical drudgery and give us more time for cultural and artistic pursuits. Machines have undoutitedly **lightened** human toil, but have also added somewhat to human misery. Modern industrialization based a minute subdivision of labor has **rebbed** the worker of the sense of beauty and craft. The craftsman has become a mechanic, a mere tool for the ends of greater production The development of intelligence and character which skillful creative work produces now sought in other activities. The workers seek their pleasure outside their work. They claim higher wages and more leisure and opportunities for recreation and relaxation. But the leisure won aftar so much effort is wasted in costly amusements. Leisure is seldom used for the noble pursuits of the mind. Besides, the industrial age has made us worshipers of wealth. In the social hierarchy, anyone who has the luck or the ability to make money occupies a high place. Before the industrial revolution, we had different standards of social valuation. Saints, men of learning, poets and philosophers were at the top of the social scale. The days when human values like honesty, sympathy, understanding justice and service were considered the most desirable qualities in man are over. Money making has become one of the most popular industries of the modern world and material progress has resulted in the creation of new wants. The consumer's appetite grows by what it feeds on. To want more and to get more is the way of material progress. By means of this exciting competition we are concealing from ourselves the barrenness of life,





The most serious consequence of the industrial movement is the passing away of the home. We see it in many western countries. Parents pursue their own pleasures with little consideration of their duties to children. The poverty of an empty life drives children into undesintile channels. Unless some **corrective** measures adopted to reverse this trend, we may see in the not so distant future a total disintegration of the family

In the context of the passage, which of the following statements is not applicable to the present-day workers?

- (a) Workers find their work more meaningful.
- (b) Workers have lost their sense of beauty.
- (c) Workers claim higher pay.
- (d) Workers have become the means for increasing production

65. Read the passage carefully and answer the following question.

Certain words/phrases in the passage are ITALICISED to help you to locate them while answering the question

The present industrial age is steadily placing into our hands power and instruments which are expected to relieve us of mechanical drudgery and give us more time for cultural and artistic pursuits. Machines have undoubtedly **lightened** human toil, but have also added somewhat to human misery. Modern industrialization based a minute subdivision of labor has **robbed** the worker of the sense of beauty and craft. The craftsman has become a mechanic, a mere tool for the ends of greater production. The development of intelligence and character which skillful creative work produces now sought in other activities. The workers seek their pleasure outside their work. They claim higher wages and more leisure and opportunities for recreation and relaxation. But the leisure won after so much effort is wasted in costly amusements. Leisure is seldom used for the noble pursuits of the mind. Besides, the industrial age has made us worshipers of wealth. In the social hierarchy, anyone who has the luck or the ability to make money occupies a high place. Before the industrial revolution, we had different standards of social valuation. Saints, men of learning, poets and philosophers were at the top of the social scale. The days when human values like honesty, sympathy, understanding justice and service were considered the most desirable qualities in man are over. Money making has become one of the most popular industries of the modern world and material progress has resulted in the creation of new wants. The

consumer's appetite grows by what it feeds on. To want more and to get more is the way of material progress. By means of this exciting competition we are concealing from ourselves the barrenness of life,

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According to the author, children go astray due to.....

- (a) Poverty of the parents
- (b) Excessive control by parents
- (c) Parents not fulfilling their obligations toward children
- (d) Peer group pressures

66. In the following question, you have to select all those options which are synonyms of the given word when the context is changed.

MEAN

- | | |
|-------------------|-----------------------------|
| (A) imply | (B) Understand |
| (C) average | (D) characterized by malice |
| (a) Only (C) | (b) Both (A) and (D) |
| (c) (A), (C), (D) | (d) (A), (B), (D) |

67. In the following question, find the word / expression which has nearly the opposite meaning of the given word.

Parsimonious

- (a) Generous
- (b) Pertaining to a religious ceremony
- (c) Hidden
- (d) Complicated

68. Pick out the most effective word from the answer words to fill in the blank to make sentence meaningfully complete.

Even in today's modern society, people still the divine to bring rain.

- | | |
|------------|----------------|
| (a) appeal | (b) provoke |
| (c) invoke | (d) propitiate |

69. In this section, each item consists of a word or phrase which is ITALICISED in the sentence given. It is followed by four words or phrase. Select the word or phrase which is closest to the opposite in meaning of the italicized word or phrase.

Let us not aggravate the sufferings of the poor.

- | | |
|------------------|---------------------|
| (a) in heart | (b) it in the heart |
| (c) by the heart | (d) no improvement |





70. What is the full form of the acronym RAM in computer systems?
- (a) Random Access Memory
(b) Remote Access Memory
(c) Read-Only Access Memory
(d) Random Arithmetic Memory
71. Which of the following is the primary elements used in computer chips?
- (a) Silicon (b) Carbon
(c) Iron (d) Uranium
72. How is data stored on a diskette?
- (a) Ink (b) Laser bubbles
(c) Magnetism (d) Circuits
73. What is the term for a temporary storage area that compensates for differences in data rate and data flow between devices?
- (a) Buffer (b) Bus
(c) Channel (d) Modem
74. What is the decimal equivalent of the binary number 11111?
- (a) 31 (b) 39
(c) 42 (d) 23
75. Which of the following is an input device?
- (a) Keyboard (b) Monitor
(c) Speaker (d) Printer
76. The first and the last sentences of the passage are numbered 1 and 6.
The rest of the passage is split into four parts and named P, Q, R, and S.
These four parts are not given in their proper order. Read the passage and find out which of the four combinations is correct
1. Proverbs contain homely but universal truths.
P. They point out the incongruities of situations in life.
Q. Naturally, therefore, they are translatable from one language to another.
R. Therefore, their appeal is direct.
S. Many of them had their birth in folk literature.
6. They are every man's philosophy.
- (a) QPRS (b) QSPR
(c) RPSQ (d) SPRQ
77. Choose the correct word to complete the sentence:
"Her approach to solving the problem was not only practical but also ____."
- (a) Dull (b) Innovative
(c) Inefficient (d) Rigid

78. Identify the error in the sentence:
"Neither the students nor the teacher were ready for the exam."
- (a) Neither (b) The students
(c) Were (d) Ready
79. Choose the most appropriate synonym for "Altruistic":
- (a) Selfish (b) Kind
(c) Greedy (d) Hostile
80. Fill in the blank:
"Her remarks were so ____ that everyone in the room was left speechless."
- (a) Concise (b) Obnoxious
(c) Eloquent (d) Vague
81. Choose the correct form of the verb:
"By the time you arrive, we ____ the meeting."
- (a) Will finish (b) Finished
(c) Will have finished (d) Will be finishing
82. Fill in the blank:
"Despite the increasing cost, the company plans to ____ with the new project."
- (a) Abandon (b) Pursue
(c) Retire (d) Halt
83. Select the correct sentence:
- (a) I would have gone to the meeting if I would have known about it
(b) I would have gone to the meeting if I had known about it.
(c) I would have gone to the meeting if I know about it.
(d) I would go to the meeting if I would have known about it.
84. Which of the following is the antonym of "Obscure"?
- (a) Vague (b) Clear
(c) Cryptic (d) Confusing
85. Choose the word that is most similar in meaning to "Nefarious":
- (a) Wicked (b) Benevolent
(c) Pious (d) Joyful
86. Fill in the blank:
"His ____ comments during the meeting upset everyone."
- (a) Tactful (b) Insensitive
(c) Delicate (d) Polite
87. The microprocessor of a computer ____
- (a) Does not understand machine language.
(b) Understands machine language and high level languages.
(c) Understands only machine language.
(d) Understands only high level languages



88. The simultaneous processing of two or more programs by multiple processors is _____
 (a) multiprogramming (b) multitasking
 (c) time-sharing (d) multiprocessing
89. What is the hexadecimal equivalent of the octal number 56?
 (a) $(2E)_{16}$ (b) $(2D)_{16}$
 (c) $(E2)_{16}$ (d) $(A2)_{16}$
90. Convert hexadecimal $(0.A35)_{16}$ to binary.
 (a) $(0.101000110101)_2$ (b) $(0.010100111010)_2$
 (c) $(101000110101)_2$ (d) $(010100111010)_2$
91. A set of rules for telling the computer what operations to perform is called a _____.
 (a) Procedural language (b) Natural language
 (c) Command language (d) Programming language
92. A(n) is a set of programs designed to manage the resources of a computer, including starting the computer, managing programs, managing memory, and coordinating tasks between input and output devices.
 (a) application suite (b) compiler
 (c) operating system (d) interface
93. What is the decimal equivalent of the binary number 10111?
 (a) 21 (b) 39 (c) 42 (d) 23
94. What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
    int y = 10000;
    int y = 34;
    printf("Hello World! %d\n", y);
    return 0;
}
```

 (a) Compile time error (b) Hello World! 34
 (c) Hello World! 1000 (d) Hello World! followed by a junk value
95. What will be the output of the following C function?

```
#include <stdio.h>
enum birds {SPARROW, PEACOCK, PARROT};
enum animals {TIGER = 8, LION, RABBIT, ZEBRA};
int main()
{
    enum birds m = TIGER;
    int k;
    k = m;
    printf("%d\n", k);
    return 0;
}
```

 (a) 0 (b) Compile time error
 (c) 1 (d) 8

96. Subtract 75 from 59 using 2's complement.
 (a) 11110000
 (b) 10100000
 (c) 10010000
 (d) 00010000
97. What will be the final values of i and j in the following C code?

```
#include <stdio.h>
int x = 0;
int f ()
{
    if (x == 0)
        return x + 1;
    else
        return x - 1;
}
int g()
{
    return x ++;
}
int main()
{
    int i = (f() + g()) || g(); //bitwise or
    printf ("%d\n", i);
    int j = g() || (f() + g()); //bitwise or
    printf ("%d\n", j);
}
```

 (a) i value is 1 and j value is 1
 (b) i value is 0 and j value is 0
 (c) i value is 1 and j value is 7
 (d) i and j value are undefined
98. How to represent -34 in one's complement form?
 (a) 00100010
 (b) 11011101
 (c) 10101010
 (d) 11110000
99. What is the smallest unit of information in a computer system?
 (a) A byte (b) A block
 (c) A bit (d) A nibble
100. Which programming technique is used to graphically represent the steps required to solve a program?
 (a) Object-oriented programming
 (b) Pseudocode
 (c) Flowchart
 (d) Top-down design





ANSWER KEY

1	2	3	4	5	6	7	8	9	10
(a)	(a)	(b)	(c)	(c)	(b)	(b)	(a)	(a)	(b)
11	12	13	14	15	16	17	18	19	20
(b)	(a)	(b)	(a)	(b)	(b)	(b)	(b)	(a)	(b)
21	22	23	24	25	26	27	28	29	30
(a)	(b)	(a)	(a)	(b)	(a)	(b)	(c)	(c)	(a)
31	32	33	34	35	36	37	38	39	40
(b)	(a)	(b)	(a)	(b)	(c)	(b)	(c)	(c)	(b)
41	42	43	44	45	46	47	48	49	50
(b)	(b)	(c)	(c)	(b)	(c)	(a)	(b)	(a)	(d)
51	52	53	54	55	56	57	58	59	60
(a)	(b)	(d)	(c)	(c)	(b)	(c)	(b)	(a)	(a)
61	62	63	64	65	66	67	68	69	70
(d)	(b)	(d)	(a)	(c)	(c)	(a)	(c)	(d)	(a)
71	72	73	74	75	76	77	78	79	80
(a)	(c)	(a)	(a)	(a)	(d)	(b)	(c)	(b)	(b)
81	82	83	84	85	86	87	88	89	90
(c)	(b)	(b)	(b)	(a)	(b)	(c)	(d)	(a)	(a)
91	92	93	94	95	96	97	98	99	100
(d)	(c)	(d)	(a)	(d)	(d)	(a)	(b)	(c)	(c)

